

Ali Abdallah

Junior AI Engineer

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Computer Science fresh graduate with 2+ years of project experience developing end-to-end machine learning and NLP solutions. Strong foundation in Object-Oriented Programming, Data Structures, Algorithms, Design Patterns, and Databases. Delivered multiple classification systems and deployed ML applications. I am highly motivated to apply Data Science, NLP, and Machine & Deep Learning techniques to real-world problems.

Education

B.Sc. in Computer Science, Ain Shams University (Class of 2024).

📅 October 2020 - July 2024

Department of **Information Systems**

📍 *Cairo, Egypt*

Grade: (B)

Graduation Project: Masha'erohom - Arabic Emotion Detector (**Grade: A+**)

Technical Skills

Programming: Python, R, C++, C#, Java, SQL (MySQL, SQLite), MongoDB, Git, Bash

Core CS: OOP, Data Structures, Algorithms, Design Patterns, System Design

Data & AI: Data Analysis, Machine Learning, Deep Learning (TensorFlow, PyTorch, Scikit-learn), NLP, Data Visualization, Statistical Analysis, Statistical Inference

Selected Projects

Masha'erohom - Arabic Emotion Detector 🖼

📅 August 2023 - July 2024

Graduation Project (Grade: A+)

- Led a team to develop an Arabic emotion detection model exceeding state-of-the-art accuracy (**92%**, **+7%** vs. prior work).
- Developed and architected an ensemble of **Bi-LSTM**, **Bi-GRU**, and **fine-tuned MARBERTv2**, stacked via **Random Forest** after extensive **hyperparameter tuning** and evaluation.
- Fine-tuned pretrained Arabic **word embeddings** to better capture dialectal nuances, diacritics, and morphology.
- Built a **Flask** web app that ingests Twitter data and visualizes public emotion distributions for user-entered keywords.
- Published a **peer-reviewed paper** 📄 following formal evaluations in IJICIS.

English Sign Language Recognition 🖼

📅 August 2023 - September 2023

- Developed a **CNN**-based model using landmark hand detection, achieving **99%** test accuracy.
- Integrated Random Forest for real-time gesture classification.

Spam Detection 🖼

📅 April 2023 - April 2023

- Built an NLP-based spam classification system using **TF-IDF** and Count Vectorization to detect unwanted SMS messages.
- Trained and evaluated multiple classifiers (**Logistic Regression**, **SVM**, **KNN**, Random Forest, **Naive Bayes**) to select the best-performing model.
- Deployed the final model as an interactive **Streamlit** web application enabling real-time predictions.

Loan Eligibility Predictor

 February 2023 - February 2023

- Developed a machine learning model to automate loan eligibility prediction based on applicant financial data.
- Performed data preprocessing and feature engineering, then trained and compared multiple classification algorithms.
- Deployed the selected **Logistic Regression** model as a user-facing **Streamlit** web application for instant eligibility assessment.

Customer Segmentation

 November 2022 - December 2022

- Performed unsupervised learning (**K-Means**, **DBSCAN**, **Hierarchical**) using R for market segmentation.
- Conducted EDA and visualized insights to support business strategy.

Relevant Experience

Machine Learning & Computer Vision Trainee

 August 2023 - September 2023

ApplAI  - Ain Shams University

- Completed intensive training in **Deep Learning and Computer Vision**, covering **CNN** fundamentals, modern architectures, object detection, and image segmentation.
- Built and evaluated multiple computer vision models using Python and deep learning frameworks.
- Achieved **1st place in 3 out of 5 Kaggle competitions**, demonstrating strong model optimization and problem-solving skills.
- Graduation project ranked in **Top 10** among all participants.

For more information about the competitions and the notebooks , the Completion Project  and the certificate .

Machine Learning Trainee

 February 2023

ApplAI  - Ain Shams University

- Gained hands-on experience with supervised and **unsupervised machine learning algorithms**.
- Applied **end-to-end ML workflows** include data preprocessing, feature engineering, model training, and evaluation.
- Delivered a complete machine learning project as part of the program.

Here is the Project and the certificate .